Evangelos Gkiolas

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EDUCATION

University of Patras, Patras, Greece 09/2020 – Present

- Diploma in Electrical and Computer Engineering (5-year Integrated Master; 300 ECTS)
  - Specialization: Systems, Control and Robotics
  - Relevant Coursework: Non-Linear Systems & Control, State-Space Linear Systems Analysis, Adaptive Control & Reinforcement Learning, Intelligent Control, Robotic Systems I, Artificial Intelligence I, Linear & Combinatorial Optimization, Applied Optimization
- Diploma Thesis (in progress): *"Puzzle Assembly via a Robotic Manipulator"*
  - Built an end-to-end vision and solving pipeline in Python that solves real 9- and 20-piece puzzles from piece shape alone
  - Segmentation: IS-Net (PyTorch / ONNX Runtime) on full-resolution crops with sub-pixel contour refinement; selected after comparing U^2-Net, IS-Net and BiRefNet (under 0.5 s per piece)
  - Shape analysis following Webster et al. (1991): skeleton-based isthmus detection of tabs and blanks, corners from the convex hull of the tab-free shape, tab/blank/flat side labelling
  - Backtracking solver (border first, then interior) that joins two sides only if they pass both an isthmus-compatibility check and a geometric check (equal length, curves agree after rigid ICP)
  - Next: backlit capture and camera calibration, then assembly with a suction-gripper robotic arm

Politecnico di Milano (PoliMi), Milan, Italy 02/2026 – 07/2026

- Erasmus+ exchange student (MSc level), Electrical Engineering
- Coursework: Advanced and Multivariable Control, Power Generation and Renewables, Smart Grids: Components, Functionalities and Benefits

PROJECTS

Optimal Football Lineup Selection – Binary Integer Programming (Python, Gurobi) 2024

- Course project: modeled starting-XI selection for Olympiacos FC (2023–24) as a BIP with 42 binary player–position variables (27-player squad) and formation/positional constraints
- Built player ratings by combining position-weighted Football Manager 2024 attributes with SofaScore match ratings; solved and compared 5 formations
- Custom sensitivity analysis for integer programs (no shadow prices): rating tolerance intervals per starter and optimal-replacement analysis for absent players
- Optimal 4-3-3 lineup matched 9 of the 11 starters fielded in the 2024 UEFA Conference League final
- Code & report: github.com/Vagkiolas/football-squad-optimization-bip

SKILLS & LANGUAGES

Programming:	Python, C, MATLAB, Java
Vision & ML:	OpenCV, scikit-image, NumPy, SciPy, PyTorch, ONNX Runtime
Tools:	Simulink, ROS, Gurobi, Git, LaTeX, LTspice, AutoCAD
Languages:	Greek (native), English (C2, fluent)